

CHANDLER PHUNG

Alabaster, AL • (626) 242-6210 • chandlerphu@gmail.com • [LinkedIn](#) • [Portfolio](#)

SUMMARY

Graduate student pursuing MS in Computer Science (December 2026) with hands-on experience building and training deep learning models (VAE, LSTM, Word2Vec) and deploying full-stack data-driven applications. Proficient in Python, PyTorch, and scikit-learn with project experience spanning NLP, computer vision, and time-series prediction across end-to-end ML pipelines.

EDUCATION

University of Alabama at Birmingham

Birmingham, AL

MS, Computer Science

Expected, December 2026

Relevant Coursework: Advanced Data Structures & Applications, Machine Learning, NLP (Natural Language Processing), GPU Programming, Deep Learning, Foundations of Data Science

BA, Computer Science

December, 2024

BS, Kinesiology with a concentration in exercise science

May, 2023

WORK EXPERIENCE

Operations Manager & Software Developer | Classic Nail Spa

January 2019 - Present

- Architected a full-stack internal dashboard (Flask REST API + React/Recharts) visualizing live POS sales, employee revenue contribution, and shop KPIs — reverse-engineered the vendor SQLite schema without documentation to expose real-time endpoints aggregating nested JSON across multiple tables
- Built an automated SMS marketing pipeline in Python: nightly cron queries the POS SQLite DB for upcoming customer birthdays, dispatches personalized messages via Twilio API, and logs delivery status — deployed as a persistent Windows service
- Deployed React SPA on a Linux Linode server with Flask backend; configured CORS, React Router for client-side navigation, and PM2 for process management

TECHNICAL PROJECTS

Song2Color - Music Genre & Tag Color Visualization

March 2026 – April 2026

- Trained Word2Vec skip-gram models on Spotify co-occurrence data producing 32-dim genre and 64-dim tag embeddings; validated cluster quality with KNN coherence and UMAP visualization
- Designed a VAE on 96-dim concatenated song vectors to learn a structured 16-dim latent space enabling smooth interpolation between songs and color generation for unseen genre-tag combinations
- Mapped embeddings to HSV color space via PCA projection (principal component angles → hue), ensuring maximally contrasting colors for dissimilar genres; resolved posterior collapse via KL floor regularization and β -annealing

F1 Lap Time Prediction - LSTM Regression Model

March 2026 – April 2026

- Built an end-to-end LSTM pipeline using FastF1 API on multi-season telemetry (speed, throttle, brake, RPM, gear, DRS) with distance-based resampling to produce fixed-length time-series inputs
- Integrated track embeddings and weather features for multi-modal prediction; implemented full ML pipeline: normalization, train/val/test splits, caching, and model checkpointing

Alarmify - Spotify Alarm Clock Web Application

December 2025 – February 2026

- Architected a full-stack web application integrating Spotify Web API with OAuth 2.0 authentication to let users create alarms that play their selected playlists at scheduled times
- Developed a Node.js/Express REST API with JWT authentication, MongoDB with Mongoose ODM, and a cron-based scheduler that triggers Spotify playback automatically
- Deployed on AWS EC2 with Application Load Balancer, Route 53, and Certificate Manager for SSL/TLS

TECHNICAL SKILLS

Programming Languages: Python, JavaScript/TypeScript, Java, C/C++, SQL, HTML/CSS

ML / AI: PyTorch, Scikit-learn, OpenCV, Word2Vec, VAE, LSTM, UMAP, PCA, Computer Vision

Data Science: Pandas, NumPy, Matplotlib, Seaborn, Jupyter, Google Colab

Web & APIs: React, Node.js, Express.js, Flask, FastAPI, Angular, RESTful APIs

Cloud & DevOps: AWS (EC2, ALB, Route 53, ACM), PM2, Git, OAuth 2.0

Databases: PostgreSQL, MongoDB, MySQL, SQLite, MongoDB Atlas